

CONVEYOR CONTROL SYSTEM – PORTFOLIO SHEET

Project: Modular Conveyor Control with Fault Latching & Batch Counting

Role: Sole designer, programmer, tester

Platform: Siemens S7-1200 | Ladder Logic

QUICK SUMMARY

Conveyor with E-stop, belt misalignment fault latching (manual reset required), batch counting with 10 sec dwell (selectable 10 or 20 products), and idle stop (no product for 10 sec → motor off).

KEY ACHIEVEMENTS

- Safety: Fault latch prevents auto-restart after belt misalignment
- Batch control: Two modes (10 or 20 products) with 10 sec dwell
- Idle stop: Auto shutdown when product flow stops
- Mode switching: Resets counter cleanly without errors

I/O COUNT: 8 Inputs | 1 Output

TEST RESULTS:

- ✓ E-stop interrupt – passes
- ✓ Fault latch + reset required – passes
- ✓ 10 product batch cycle – passes
- ✓ 20 product batch cycle – passes
- ✓ Mode switch mid-cycle – counter resets correctly
- ✓ Idle stop (no product 10 sec) – passes

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GitHub: <https://github.com/afnanjawad/PLC-Conveyor-Control>

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